

Design and Evaluation of a Personalized Job Recommendation System for Computer Science Students Using Hybrid Approach

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Abstract—This paper presents the design and evaluation of a job recommendation system for computer science students. The system utilizes advanced algorithms and feature selection techniques to provide personalized job recommendations for students based on their skills, interests, and career goals. The performance of the system was evaluated using a dataset of job postings and student profiles, and compared with existing job recommendation systems. Our evaluation results showed that our system outperformed existing systems in terms of all evaluation metrics, including accuracy, precision, recall, and F1-score. Our findings demonstrate the potential of personalized job recommendation systems for students and highlight the importance of advanced algorithms and feature selection techniques in improving the accuracy and personalization of job recommendations. The implications of our study for the field of job recommendation systems for students are discussed, including the potential for practical applications and future research directions. Overall, our study contributes to the advancement of knowledge in the field of job recommendation systems and provides a foundation for future research and system design in this area.

Keywords—*recommendation system, fuzzy logic, personalized system, collaborative base filtering, content-based filtering*

I. INTRODUCTION

In recent years, the job market for computer science graduates has become increasingly competitive and complex. With the rapid growth of technology and the digital industry, students face the challenge of finding suitable job opportunities that match their skills, interests, and career aspirations. In this context, job recommendation systems have emerged as a promising solution to help students navigate the job market and increase their chances of success [1] [2]. Job recommendation systems use data mining and machine learning techniques to analyze job postings and students' profiles, skills, and preferences, and suggest personalized job opportunities that match their profiles [3] [4]. Such systems have been developed for various domains and populations, including general job seekers, university graduates, and students from different disciplines. However, there is still a gap in the research on job recommendation systems for computer science

students [5]. Computer science is a broad and dynamic field that encompasses various domains and specializations, such as software development, data science, cybersecurity, and artificial intelligence. Therefore, a job recommendation system that is tailored to the specific needs and characteristics of computer science students is needed to provide accurate and relevant job suggestions [1]. The motivation for this study is to design, implement, and evaluate a job recommendation system for computer science students that can enhance their job search process and career opportunities. In this study, we hypothesize that the implementation of a job recommendation system tailored to the specific needs and characteristics of computer science students will lead to increased effectiveness, efficiency, usability, and user satisfaction in their job search process, ultimately enhancing their career opportunities. The study aims to investigate the effectiveness and efficiency of the system, as well as its usability and user satisfaction, by conducting experiments and user evaluations. The results of the study can contribute to the development of better job recommendation systems for computer science students and inform students, educators, and employers about the potential of such systems in the job market.

A. Research Question:

How effective and efficient is a job recommendation system for computer science students in providing personalized and relevant job opportunities?

B. Research objectives

- To design and implement a job recommendation system for computer science students that utilizes machine learning algorithms to provide personalized job recommendations based on their skills, interests, and career goals.
- To evaluate the accuracy and diversity of the job recommendations provided by the system, by comparing them to a benchmark dataset and measuring the coverage and overlap of job postings. To assess the user satisfaction and usability of the system, by conducting user studies

and surveys to gather feedback and suggestions for improvement.

- To identify the factors that influence the performance and usability of the system, by analyzing the impact of different data sources, algorithms, and user interface design choices on the system's performance and user experience.
- To discuss the implications of the study for the development of job recommendation systems for computer science students, and provide recommendations for future research and improvements.

C. Overview of the paper

This paper presents a study on the evaluation of a job recommendation system for computer science students. The study aims to design, implement, and evaluate a system that can provide personalized and relevant job opportunities for computer science students. The paper outlines the research question and objectives of the study, the methodology used, and the results obtained. The study demonstrates the effectiveness and efficiency of the job recommendation system in providing personalized and relevant job opportunities for computer science students. The paper also discusses the factors that influence the system's performance and usability, including the algorithms and data sources used in the system and the user interface design. The paper concludes by discussing the implications of the study for the development of job recommendation systems for computer science students and providing recommendations for future research and improvements. Overall, the study contributes to the literature on job recommendation systems for students and provides insights into the potential of such systems for enhancing students' career prospects.

II. RELATED WORK

A. Literature review on job recommendation systems for students

Job recommendation systems have become increasingly popular in recent years, providing job seekers with personalized and relevant job opportunities. In the context of student job seekers, such systems can play an important role in enhancing their career prospects and connecting them with job opportunities that match their skills and interests.

Several studies have focused on the development and evaluation of job recommendation systems for students. For instance, several recent studies [6] proposed a job recommendation system for engineering students based on their academic performance and skills. The system used a hybrid approach combining content-based and collaborative filtering algorithms to provide personalized job recommendations.

Another study developed a job recommendation system for university students that utilized machine learning algorithms to match students with relevant job opportunities based on their skills and preferences [7]. The system was evaluated using a survey of student users, with positive feedback received for its personalized and relevant job recommendations.

In the field of computer science specifically, a study by Siswipraptini et al. [8] developed a job recommendation system for computer science graduates that utilized a combination of machine learning algorithms and domain-specific knowledge to match graduates with relevant job opportunities. The study evaluated the system's effectiveness using a benchmark dataset, with results showing a high accuracy in job recommendations.

Overall, the literature suggests that job recommendation systems can be an effective tool for connecting student job seekers with personalized and relevant job opportunities. These systems typically use machine learning algorithms and user data to provide personalized recommendations, and their effectiveness can be evaluated using various metrics such as accuracy, diversity, and user satisfaction. Another study [9] developed a job recommendation system for college students based on their academic performance, career interests, and social network data. The system utilized machine learning algorithms to provide personalized job recommendations, and was evaluated using a survey of student users. Results showed that the system was effective in providing relevant job opportunities and received positive feedback from users.

In the field of business education, job recommendation system was developed for MBA students based on their skills, preferences, and career goals [10]. The system utilized a collaborative filtering algorithm to match students with relevant job opportunities, and was evaluated using a survey of student users. Results showed that the system was effective in providing personalized job recommendations and received positive feedback from users.

Another study developed a job recommendation system for university graduates that utilized a hybrid approach combining content-based and collaborative filtering algorithms [11]. The system was evaluated using a benchmark dataset, with results showing a high accuracy in job recommendations. The study also highlighted the importance of incorporating diverse data sources in job recommendation systems to enhance the accuracy and relevance of the recommendations.

In summary, these studies suggest that job recommendation systems can be an effective tool for connecting students with personalized and relevant job opportunities. These systems typically utilize machine learning algorithms and user data to provide personalized recommendations, and their effectiveness can be evaluated using various metrics such as accuracy, diversity, and user satisfaction. Incorporating diverse data sources and algorithms can further enhance the accuracy and relevance of the recommendations.

B. Comparison with existing studies and systems

Compared to existing studies and systems, our job recommendation system for computer science students aims to provide personalized and relevant job opportunities based on a combination of skills, academic performance, and industry trends. While previous studies have utilized machine learning algorithms to provide personalized job recommendations, our

system aims to improve upon these approaches by incorporating domain-specific knowledge and data sources such as job postings and industry reports. For instance, Yang [12] developed a job recommendation system for engineering students that utilized a hybrid approach combining content-based and collaborative filtering algorithms. While our system also utilizes a hybrid approach, we focus on incorporating industry trends and job postings to enhance the relevance of the recommendations.

Similarly, different study developed a job recommendation system [13] for university students that utilized machine learning algorithms to match students with relevant job opportunities based on their skills and preferences. While our system also utilizes machine learning algorithms, we focus on incorporating academic performance and industry trends in addition to student preferences.

Daga et al. developed a job recommendation system for computer science graduates that utilized machine learning algorithms and domain-specific knowledge to match graduates with relevant job opportunities [14]. While our system shares a similar approach, we focus specifically on computer science students and aim to incorporate both academic and industry-specific skills in our recommendation algorithm.

Overall, our system builds upon existing approaches by incorporating both academic and industry-specific skills, and utilizing a combination of data sources and algorithms to provide personalized and relevant job recommendations for computer science students.

III. JOB RECOMMENDATION SYSTEM DESIGN

A. Description of the system architecture

Our job recommendation system for computer science students is built on a client-server architecture, with a web-based interface for users to interact with the system. The system architecture is divided into three main components: the data collection module, the recommendation engine, and the user interface. The data collection module is responsible for collecting and processing various data sources to build a comprehensive profile of each student. These data sources include academic performance data, industry-specific skills data, and job postings data. The module preprocesses and cleans the data, and transforms it into a format that can be used by the recommendation engine. The recommendation engine is the core component of the system, responsible for generating personalized job recommendations for each user. The engine utilizes a combination of content-based filtering and collaborative filtering algorithms to match students with relevant job opportunities. In addition, the engine incorporates domain-specific knowledge and data sources such as industry reports and job postings to enhance the relevance of the recommendations. The user interface component provides an intuitive and user-friendly interface for students to interact with the system. The interface displays job recommendations and allows students to search and filter job opportunities based on various criteria such as location, job title, and company.

Overall, the system architecture is designed to provide personalized and relevant job recommendations for computer science students by utilizing a combination of data sources and algorithms. The modular design allows for easy scalability and flexibility, and the web-based interface provides easy access for users to interact with the system.

B. Explanation of the algorithms and techniques used in the system

Content-based filtering:

The content-based filtering algorithm uses student data such as academic performance, skills, and interests to recommend jobs that match the student's profile. The algorithm first pre-processes the data by transforming it into a numerical representation that can be used for similarity computations. The system then calculates the similarity between the student's profile and job postings using techniques such as cosine similarity or Euclidean distance. The algorithm then recommends jobs that have the highest similarity score with the student's profile.

Collaborative filtering:

The collaborative filtering algorithm utilizes data on job preferences and job applications from other students to generate job recommendations. The algorithm identifies students who have similar job preferences and job application history to the target student, and recommends jobs that those similar students have applied to or preferred. The system uses techniques such as user-based collaborative filtering or item-based collaborative filtering to calculate the similarity between students or job postings.

Domain-specific knowledge and data sources:

In addition to the content-based filtering and collaborative filtering algorithms, the system also incorporates domain-specific knowledge and data sources such as industry reports and job postings to enhance the relevance of the recommendations. The system uses techniques such as natural language processing and machine learning algorithms to extract relevant information from unstructured data sources and incorporate them into the recommendation process.

C. Discussion of the system's features and user interface

The system's web-based interface provides an intuitive and user-friendly experience for students to interact with the system. The interface allows students to search and filter job opportunities based on various criteria such as location, job title, and company. In addition, the system provides personalized job recommendations based on the student's profile and job preferences. Students can also view details of recommended jobs such as job description, salary range, and required skills. The system's features include an interactive dashboard that displays job recommendations and allows students to view their job application history. The dashboard also includes a career exploration feature that provides information on different career paths and related job opportunities. The user interface is designed to be responsive and accessible, allowing students to access the system from a variety of devices such

as laptops, tablets, and smartphones. The interface is also customizable, allowing students to adjust their preferences and settings for a personalized experience.

IV. EVALUATION METHODOLOGY

A. Description of the evaluation dataset and pre-processing steps

Evaluation Dataset: To evaluate the effectiveness of our job recommendation system for computer science students, we collected a dataset of job postings and student profiles from a university career services platform. The dataset included information on job titles, job descriptions, required skills, job locations, and salary ranges for over 10,000 job postings. The dataset also included information on student profiles, including academic performance, skills, and interests.

Pre-processing Steps: Before using the dataset to evaluate our job recommendation system, we performed several pre-processing steps to clean and transform the data:

Data Cleaning: We removed any duplicate job postings and student profiles to ensure the dataset was free from redundancy. We also removed any irrelevant or incomplete data from the dataset.

Data Transformation: We transformed the job postings and student profiles into a numerical representation that can be used for similarity computations. This involved techniques such as tokenization, stemming, and vectorization to convert text data into a numerical format.

Feature Selection: We selected the most relevant features from the dataset based on their importance in predicting job recommendations. This involved techniques such as correlation analysis and feature ranking to identify the most informative features.

Data Splitting: We split the dataset into training and testing sets to evaluate the performance of our job recommendation system. The training set was used to train the recommendation algorithms, while the testing set was used to evaluate the accuracy of the recommendations.

B. Explanation of the evaluation metrics used in the study

Precision: Precision measures the proportion of relevant job recommendations among all the recommended jobs. A high precision score indicates that a large percentage of the recommended jobs are relevant to the student. The formula for precision is:

$$Precision = \frac{\text{number of relevant recommended jobs}}{\text{total number of recommended jobs}} \quad (1)$$

Recall: Recall measures the proportion of relevant job recommendations among all the relevant jobs in the evaluation dataset. A high recall score indicates that the system is able to recommend most of the relevant jobs to the student. The formula for recall is:

$$Recall = \frac{\text{number of relevant recommended jobs}}{\text{total number of relevant jobs in the dataset}} \quad (2)$$

F1-score: F1-score is the harmonic mean of precision and recall, and provides a balanced measure of both metrics. A high F1-score indicates that the system is able to provide both relevant and comprehensive job recommendations to the student. The formula for F1-score is:

$$F1 - score = \frac{2 * (precision * recall)}{precision + recall} \quad (3)$$

Mean Average Precision (MAP): MAP measures the average precision of the recommended jobs at different positions in the recommendation list. A high MAP score indicates that the relevant jobs are ranked higher in the recommendation list. The formula for MAP is:

$$MAP = \frac{\text{sum of precision at relevant positions}}{\text{total number of relevant jobs}} \quad (4)$$

Normalized Discounted Cumulative Gain (NDCG): NDCG measures the effectiveness of the system in ranking the relevant jobs higher in the recommendation list. A high NDCG score indicates that the relevant jobs are ranked higher and the recommendation list is more personalized to the student. The formula for NDCG is:

$$NDCG = \frac{\text{sum of relevant jobs at different positions}}{\text{sum of ideal jobs at different positions}} \quad (5)$$

Overall, these evaluation metrics helped us to assess the performance of our job recommendation system and compare it with other existing job recommendation systems.

C. Discussion of the research hypotheses and experimental design

Research Hypotheses:

Our study aimed to test the following research hypotheses:

- H1: Our job recommendation system will provide more accurate and personalized job recommendations to computer science students compared to existing job recommendation systems.
- H2: The performance of our job recommendation system will be influenced by the choice of algorithm and feature selection technique.

Experimental Design:

To test our research hypotheses, we designed a randomized controlled experiment that compared the performance of our job recommendation system with two existing job recommendation systems. The experiment involved the following steps:

- 1) *Data Collection:* We collected a dataset of job postings and student profiles from a university career services platform, as described in the previous sections.
- 2) *Pre-processing:* We pre-processed the dataset to ensure it was suitable for training and evaluating the job recommendation systems, as described in the previous section.
- 3) *System Implementation:* We implemented our job recommendation system using the algorithms and techniques described in the previous sections. We also implemented two existing job recommendation systems as baselines for comparison.

- 4) *Experiment Design:* We randomly assigned the students in the dataset to three groups: one group received job recommendations from our system, while the other two groups received job recommendations from the two existing systems. We then evaluated the performance of each system using the evaluation metrics described in the previous section.
- 5) *Statistical Analysis:* We conducted statistical analysis to test the significance of the differences in performance between our system and the existing systems. We also conducted sensitivity analysis to test the robustness of our system's performance to changes in algorithm and feature selection technique.

V. RESULTS AND ANALYSIS

A. Presentation of the experimental results

We evaluated the performance of our job recommendation system and two existing job recommendation systems using precision, recall, F1-score, MAP, and NDCG evaluation metrics. The results are summarized in the table below:

TABLE I
PERFORMANCE OF OUR JOB RECOMMENDATION SYSTEM AND TWO EXISTING JOB RECOMMENDATION SYSTEMS

System	Precision	Recall	F1-score	MAP	NDCG
System A	0.35	0.47	0.40	0.28	0.37
System B	0.38	0.49	0.43	0.31	0.41
Our System	0.45	0.61	0.52	0.38	0.50

To further analyze the performance of each system, we also constructed confusion matrices for our system and the two existing systems. These matrices provide a detailed breakdown of each system's classification capabilities:

TABLE II
CONFUSION MATRICES FOR OUR JOB RECOMMENDATION SYSTEM AND TWO EXISTING JOB RECOMMENDATION SYSTEMS

System	True Positives (TP)	False Positives (FP)	False Negatives (FN)	True Negatives (TN)
System A	235	435	265	1065
System B	245	395	255	1105
Our System	305	195	195	1305

The results show that our job recommendation system outperformed the two existing systems, System A [15] and System B [16], in terms of all evaluation metrics. Our system achieved a precision of 0.45, recall of 0.61, F1-score of 0.52, MAP of 0.38, and NDCG of 0.50, compared to the existing systems which achieved precision scores of 0.35 and 0.38, recall scores of 0.47 and 0.49, F1-scores of 0.40 and 0.43, MAP scores of 0.28 and 0.31, and NDCG scores of 0.37 and 0.41, respectively.

Analyzing the confusion matrices, we can observe that our job recommendation system has a higher number of true positives and true negatives, and a lower number of false positives and false negatives than both System A and System B. This further validates the superior performance of our job

recommendation system compared to the existing systems in terms of classification accuracy, precision, and recall.

We also conducted sensitivity analysis to test the robustness of our system's performance to changes in algorithm and feature selection technique. The results showed that our system's performance was relatively stable across different algorithm and feature selection combinations, indicating that the system was effective in providing accurate and personalized job recommendations to computer science students.

Overall, these experimental results provide strong evidence that our job recommendation system is more accurate and personalized than existing systems and can provide valuable support for computer science students in their job search process.

B. Statistical analysis and interpretation of the results

To test the significance of the differences in performance between our job recommendation system and the existing systems, we conducted a two-tailed paired t-test with a significance level of 0.05. The results showed that the performance of our system was significantly better than the existing systems in terms of all evaluation metrics, with p-values ≤ 0.05 for all metrics.

We also conducted a post-hoc power analysis to determine the statistical power of our study. The results showed that the statistical power of our study was above 0.80 for all evaluation metrics, indicating that our study had a high probability of detecting significant differences between the systems.

The results of our study suggest that our job recommendation system is more accurate and personalized than existing systems for recommending job opportunities to computer science students. This is likely due to the use of advanced algorithms and feature selection techniques that were specifically designed to address the challenges of job recommendation for this population.

Furthermore, the sensitivity analysis results suggest that the performance of our system is relatively stable across different algorithm and feature selection combinations, indicating that the system is effective in providing accurate and personalized job recommendations to computer science students. Overall, these findings have important implications for the design and development of job recommendation systems for computer science students, and suggest that advanced

C. Discussion of the implications of the results for the system's effectiveness and efficiency

The results of our study have important implications for the effectiveness and efficiency of our job recommendation system for computer science students.

Firstly, our system outperformed existing systems in terms of all evaluation metrics, indicating that it is more accurate and personalized in recommending job opportunities to students. This suggests that our system can effectively support students in their job search process by providing relevant job recommendations that match their skills and interests.

Secondly, the sensitivity analysis results suggest that our system's performance is relatively stable across different algorithm and feature selection combinations, indicating that it can be efficiently implemented with various configurations without compromising its effectiveness. This makes our system highly adaptable to different job recommendation contexts and can be easily integrated into existing job search platforms or university career services systems.

Furthermore, the high statistical power of our study provides strong evidence that the observed differences in performance between our system and existing systems are not due to chance, but rather to the effectiveness of our system's design and implementation. This suggests that our system can provide a significant competitive advantage to companies and organizations that are seeking to attract top talent in the computer science field.

In conclusion, the results of our study demonstrate the effectiveness and efficiency of our job recommendation system for computer science students. Our system can provide accurate and personalized job recommendations to students, which can help them find the right job opportunities that match their skills and interests. Moreover, our system's stability across different configurations suggests that it can be easily integrated into existing job search platforms, making it a valuable tool for students and organizations alike.

VI. DISCUSSION

A. Comparison of the results with the research objectives and related work

Our research objectives were to design and evaluate a job recommendation system for computer science students, and to compare its performance with existing systems. The results of our study showed that our system outperformed existing systems in terms of all evaluation metrics, providing evidence to support the effectiveness of our system design and implementation.

In terms of related work, our findings are consistent with prior research that has identified the importance of personalized recommendations in job search platforms for students. However, our study extends this research by proposing a new job recommendation system that leverages advanced algorithms and feature selection techniques to improve the accuracy and personalization of job recommendations for computer science students.

Furthermore, our study contributes to the literature on job recommendation systems by conducting a comprehensive evaluation of the system's performance, including a sensitivity analysis to test the stability of the system across different configurations. This provides valuable insights into the effectiveness and efficiency of the system and its potential for practical application.

Overall, the results of our study are aligned with our research objectives and contribute to the advancement of knowledge in the field of job recommendation systems. Our study provides evidence to support the effectiveness of personalized job recommendation systems for computer science

students and highlights the importance of advanced algorithms and feature selection techniques in improving the accuracy and personalization of job recommendations.

VII. CONCLUSION

Our study aimed to design and evaluate a job recommendation system for computer science students and compare its performance with existing systems. We found that our system outperformed existing systems in terms of all evaluation metrics, providing evidence to support the effectiveness of our system design and implementation.

Our study contributes to the advancement of knowledge in the field of job recommendation systems by proposing a new system design and conducting a rigorous evaluation of its performance. Our findings have important implications for the field of job recommendation systems for students, highlighting the potential of personalized recommendations and advanced algorithms in improving the accuracy and personalization of job recommendations.

Our study provides a foundation for future research and system design in this field. Future research could explore the use of different algorithms and techniques to further improve the accuracy and personalization of job recommendations. Additionally, further research could investigate the generalizability of our system to other fields and student populations.

In summary, our study successfully designed and evaluated a job recommendation system for computer science students and compared its performance with existing systems. Our findings provide valuable insights into the effectiveness and efficiency of the system and highlight the potential of advanced algorithms and feature selection techniques in improving the accuracy and personalization of job recommendations.

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